Generative genomics accurately predicts future experimental results

Synthesize Bio

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Abstract

AI models capable of predicting experimental outcomes could accelerate biomedical research by circumventing fundamental constraints of laboratory experimentation and clinical trials. We developed GEM-1 (Generate Expression Model-1), a generative genomics framework that models the diversity of real-world gene expression experiments and accurately predicts future experimental results.

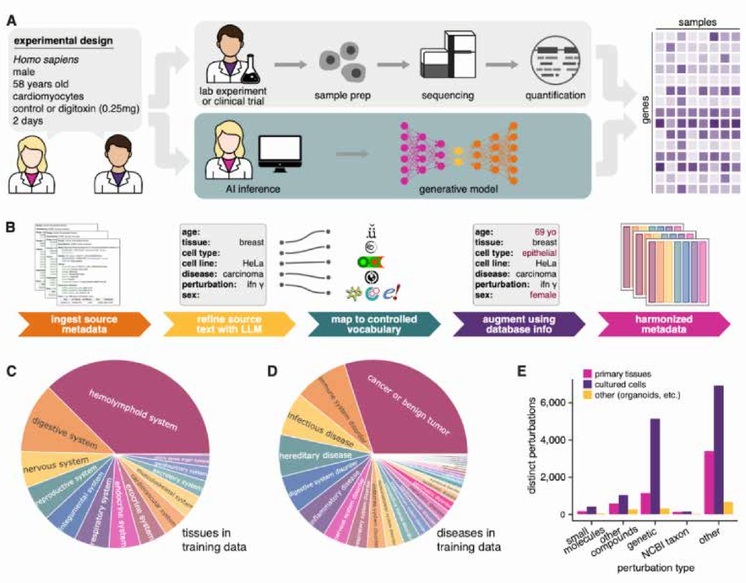
We trained GEM-1 using 470,691 bulk RNA-seq samples from 24,715 datasets in the NCBI Sequence Read Archive, spanning diverse tissues, diseases, and over 18,000 distinct perturbations. An automated metadata agent harmonized fragmented experimental descriptions using large language models. GEM-1 employs a deep latent variable model that partitions experimental metadata into biological, technical, and perturbational components, using pretrained foundation model embeddings to enable generalization to novel perturbations.

Testing on holdout data deposited after training, GEM-1 achieved pseudoreplicate-level accuracy (pearson correlation of gene rank across samples, r_gene, of 0.65-0.75) for previously observed contexts and maintained strong performance for completely novel genetic (r_gene = 0.58-0.63) and chemical perturbations (r_gene = 0.52-0.68). We extended GEM-1 to single-cell data using 41.5 million cells, achieving comparable performance to established models for cell type annotation while enabling interpretable biological feature inference.

We demonstrated clinical utility by generating synthetic cohorts that accurately recapitulated key biological phenomena in SLE (200 samples correctly modeling lupus interferon dysregulation and cancer (10,523 samples exhibiting known molecular features of cancer).

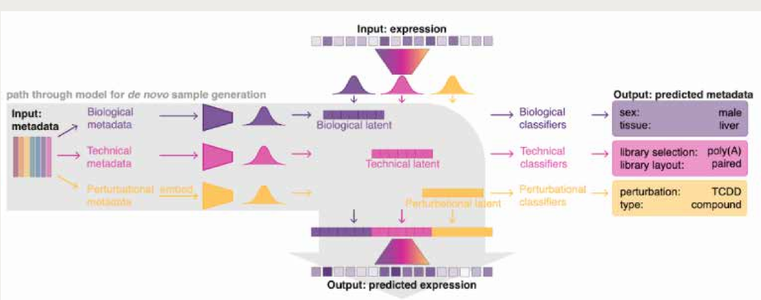
This approach represents a significant advance toward AI systems that can predict experimental outcomes before physical experiments are conducted, shortcutting fundamental limitations in experimental speed and clinical trial recruitment and potentially revolutionizing drug development and personalized medicine.

Methods



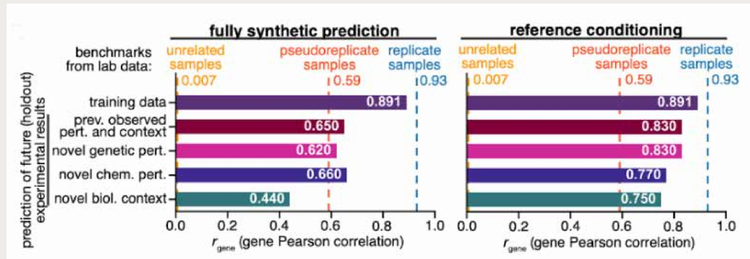
GEM-1 is a generative genomics framework that models real-world gene expression experiments and accurately predicts future experimental results.

A. GEM-1 mirrors the process of a scientist conducting a laboratory experiment or clinical trial and measuring resulting gene expression.
B. The metadata agent processes and harmonizes source metadata.
C & D. Bulk RNA-seq training data represents a diversity of tissue types and diseases.
E. GEM-1 models a variety of distinct perturbations.

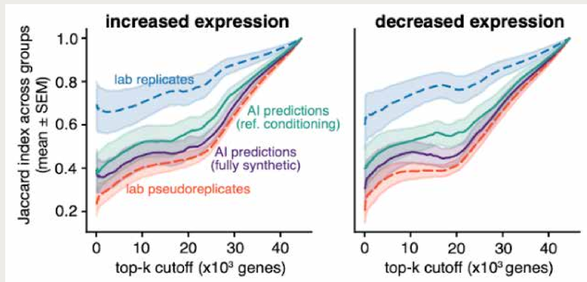


GEM-1's architecture and path through the model for de novo sample generation.

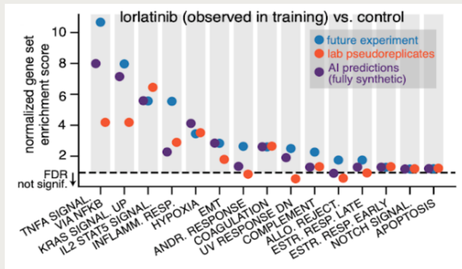
AI Bulk Expression Data is Like Lab Data



The GEM-1 model performs well at predicting gene expression based on simple input prompts about a sample (left), and even better when predicting how a perturbation will alter a specified gene expression profile (right). The five bars are the training data and each of four holdout datasets. Benchmarks: "unrelated samples" are randomly chosen lab samples; "pseudoreplicate samples," are with matched metadata from different studies; "replicate samples," are with matched metadata in the same study.

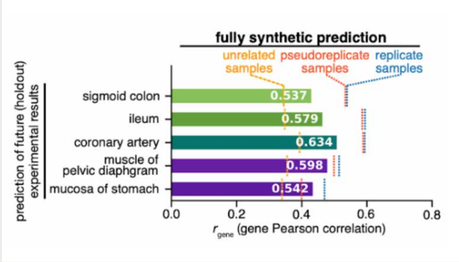


AI data differential expression surpasses between-study lab data. Jaccard index (size of intersection divided by size of union) of the top k differentially expressed genes (x axis) shows that fully synthetic AI predictions yield more consistent results than between-study (i.e. pseudoreplicate) contrasts. Analysis was performed across all studies in the holdout sets for which pseudoreplicate samples could be identified.

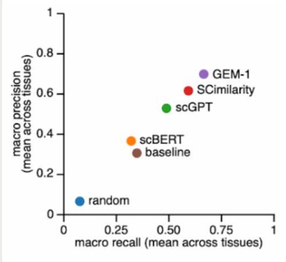


AI data identifies biological signals following drug treatment. Gene set enrichment following lorlatinib treatment of NSCLC cells in a holdout study (BioProject PRJEB53629), a pseudoreplicate study (BioProject PRJEB50324), and AI data. Sets shown are the most enriched Hallmark Pathways from MSigDB, identified using a hypergeometric test on log fold-change values from the holdout study.

AI Single-cell Expression Data is Like Lab Data

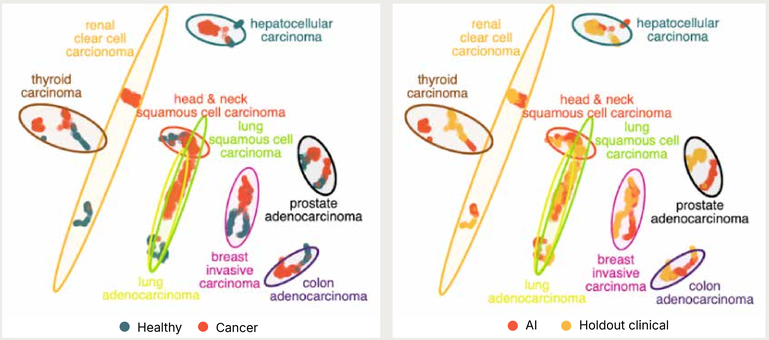


Model performance for holdout data from the Tabula Sapiens dataset. "Unrelated samples," are randomly chosen samples; "pseudoreplicate samples," are samples from the same tissue and cell type, but from different donors; "replicate samples," are samples from the same tissue, cell type, and donors.

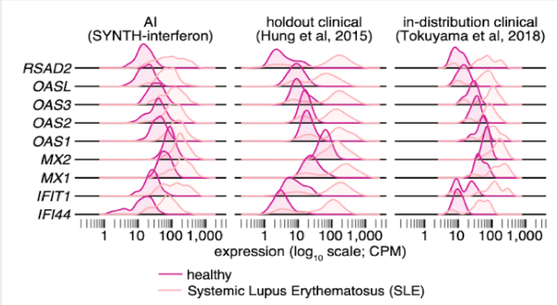


Precision and recall for scRNAseq cell type annotation with a linear probe, averaged across all tissues in the holdout data.

AI-generated Samples Look Real



UMAP dimension reduction showing similarity of AI cancer and holdout clinical (TCGA) data. We generated expression data from 27 cancer types, as well as normal control samples when relevant, for a total of 10,523 samples. We designed this AI generated cancer dataset to mimic the structure of The Cancer Genome Atlas (TCGA), which we excluded from our training data.

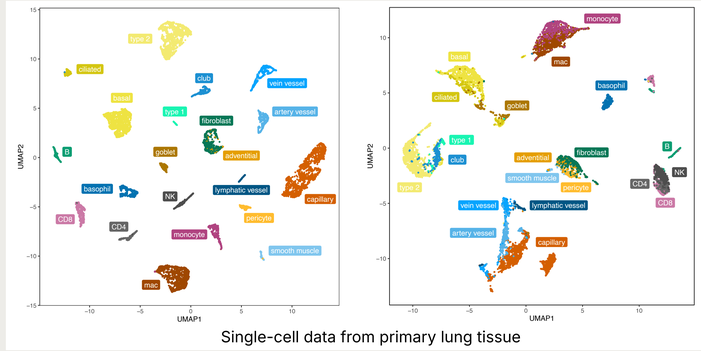


Expression of IFN induced genes are upmodulated in AI-generated SLE vs. healthy samples as in clinical lab data. We generated expression data corresponding to blood samples taken from a cohort of individuals who were healthy (n = 100) or diagnosed with an autoimmune disease, Systemic Lupus Erythematosus (SLE; n = 100). We compare this AI generated interferon dataset to data from two clinical SLE cohorts, one held out and one observed during training.

Conclusions

- GEM-1 predicts future gene expression experiments with accuracy matching lab-level pseudoreplicate reproducibility, including for novel perturbations.
- The model's interpretable architecture with separable latent factors enables flexible in silico experimental strategies such as reference conditioning.
- GEM-1 performs comparably to specialized foundation models on both bulk and single-cell RNA-seq annotation and generation tasks.
- This generative genomics framework demonstrates the feasibility of computationally simulating gene expression results to accelerate biomedical research.

Can you tell which one is AI and which is lab data?



Single-cell data from primary lung tissue

We turned our goal for biological realism into a game of trying to figure out if parallel datasets come from the lab or our AI model. Try playing our game and see how you do! →